IoT in Healthcare: Applications, Challenges and Future Concerns

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***Abstract*—The Internet of Things (IoT) is rapidly transforming healthcare, promising personalized care and reducing strain on systems. With the advancement of interconnected systems, it's possible to imagine devices whispering updates to doctors, pills reminding the patients to take, and remote monitoring for the elderly. In the developing region, the rise of IoT arises from several factors like; limited medical access, rising chronic diseases, expensive healthcare, and the growing demand for telemedicine. To leverage this potential, researchers are working in this area. The approaches have been classified into five categories; sensors gathering data, managing resources efficiently, ensuring smooth communication, building useful applications, and prioritizing security. The thorough analysis shed light on the benefits and limitations of each approach, along with their evaluation techniques, tools, and metrics. Further, Energy management, privacy concerns, fog computing, and resource allocation remain open challenges. To be able to make correct predictions, Machine Learning (ML) algorithms must be woven into the Health-Internet of Things (H-IoT) given that healthcare deals with an enormous volume of data. However, hurdles like interoperability, real-world implementation, scalability, and device mobility need to be addressed for successful integration. The paper explores how IoT stands poised to revolutionize healthcare. By tackling the existing challenges and capitalizing on emerging trends, one can unlock a future where technology empowers personalized care, improves the quality of life, and makes healthcare accessible to all.**

***Keywords*— Healthcare, Sensors, IoT, H-IoT, Machine Learning**

I. Introduction

The healthcare industry has undergone a revolution in recent years due to the rapid advancements in technology. Gone are the days of solely hospital-based diagnoses and lengthy in-patient treatments. Miniaturized devices like smartwatches now enable convenient diagnosis and health monitoring at home, empowering patients and shifting the focus to a patient-centric model. Clinical analyses for things like blood pressure, glucose levels, and oxygen saturation can be done without healthcare professionals present, and with advanced telecommunication services, even remote areas can connect their clinical data to healthcare centers [1]. These advancements, coupled with cutting-edge technologies like machine learning, big data analysis, and the IoT, have drastically improved the accessibility of healthcare facilities, transforming the entire landscape of healthcare delivery [2]. Fig. 1 shows the techniques of IoT in healthcare.

The explosion of real-time data from wearables and other H-IoT devices presents a vast, but complex opportunity for healthcare. ML algorithms are crucial in unlocking this potential, extracting valuable insights, and enabling life-changing inferences. However, achieving high accuracy with these models isn't a simple feat. Current research focuses on discovering new ML applications for H-IoT, assessing their suitability, and continuously pushing the boundaries of predictive and analytical accuracy. In simpler terms, researchers are working tirelessly to refine how we translate H-IoT data into actionable insights that improve healthcare outcomes [3][18].

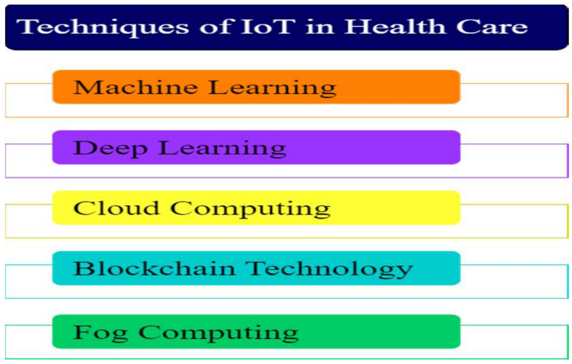

Fig. 1. IoT Techniques in Healthcare

Deep learning now mines a vast treasure trove of information - from electronic health records and medical images to mobile app data and patient-generated wearables. Armed with this diverse data, algorithms can predict diseases, assist in diagnoses, and even guide clinical decision-making. Beyond the hospital walls, deep learning's tentacles reach even deeper. In the realm of biomedicine and pharmaceuticals, it powers breakthroughs in molecular diagnostics, pharmacogenomics, and personalized cancer care. From pinpointing pathogenic variants to revolutionizing drug discovery, deep learning is changing the very fabric of medical research [4][19].

Sensor devices and IoT networks will become commonplace, empowering institutions to implement effective health management solutions. For this vision to come alive, governments and policymakers must craft a robust strategy encompassing intelligent devices, technology, sensors, and IoT systems. A holistic approach is crucial, drawing expertise from diverse fields like engineering, computer science, architecture, and, most importantly, healthcare. These initiatives offer a glimpse into a future where technology seamlessly blends with our lives, enhancing health and ushering in a new era of human-centric urban experience [20][21].

Fog computing revolutionizes healthcare by shifting data processing closer to the source. Instead of solely relying on distant cloud servers or centralized healthcare centers, fog computing empowers sensors and nearby devices to analyze measured data directly. This decentralization unlocks the potential for faster diagnoses and real-time alerts, thanks to reduced data transmission delays. Moreover, unlike cloud solutions concentrated in large remote facilities, fog

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computing offers localized computational power, enabling healthcare organizations to tailor data processing and services to their specific needs and resources. This empowers them to manage medical data efficiently and flexibly, while still benefiting from the broader capabilities of cloud services when needed [5][22[24].

II. Literature Review

This section presents the researcher's work in IoT healthcare. Table 1 presents the summary of the proposed method used by researchers. Zhao et al. [6] conducted a comprehensive review of deep learning research in machine health monitoring systems (MHMS). MHMS systems leverage deep neural networks with multi-layered nonlinearities to extract progressively meaningful insights from health data. The authors categorize existing DL-based MHMS based on four architectural variations: Autoencoders, Restricted Boltzmann Machines, Convolutional Neural Networks, and Recurrent Neural Networks. Costa et al. [7] investigated the potential of the Internet of Health Things (IoHT) in healthcare. This network aims to improve patient care by seamlessly acquiring and merging critical vital sign data within hospital settings. Through four key stages - collection, storage, processing, and presentation - IoHT empowers more informed clinical decision-making and resource allocation. While promising, the authors acknowledge two main challenges associated with IoHT; Firstly, the sheer volume and complexity of collected data can overwhelm caregivers, potentially hindering its effective utilization. Secondly, the switching of medical devices within the network adds another layer of complexity to managing and interpreting the data accurately.

Riazul Islam et al. [8] explored the diverse capabilities of IoT in healthcare, proposing various medical network structures and platforms to facilitate data collection and transfer. They identified two key benefits of implementing these services: 1) cost reduction in healthcare delivery, and 2) improved quality of life for patients. However, a key challenge lies in the limitations of currently deployed IoT devices. Many rely on relatively slow processors and lack the computational power to handle complex tasks, potentially hindering their full potential in the field.

D Souza et al. [9] offer a novel solution for indoor patient tracking and monitoring in healthcare settings. Their system utilizes a network of static nodes strategically placed throughout a hospital facility. These nodes interact with portable units carried by patients, utilizing Fleck Nano wireless sensors and accelerometers to determine location and physical activity (e.g., walking, running). While effective in controlled environments, relying solely on mobile device GPS and Wi-Fi for location tracking raises concerns about battery life due to their significant power consumption. While Kadhim et al. [10] provided a valuable overview of Internet-based healthcare monitoring systems and explored the potential of IoT applications in enhancing traditional medical care, their review suffers from some limitations. They primarily utilized a descriptive approach, lacking a systematic review methodology. This raises concerns about potential selection bias and the generalizability of their findings. Additionally, the paper fails to specify the taxonomy used for classifying reviewed studies, the number and range of years covered in the analysis, and any plans for future research directions. Strengthening these aspects would significantly improve the robustness and impact of their work.

Table I. Researchers Work

Author Name/Proposed Method	Summary
Zhao et al. [6]/ Machine Health Monitoring Systems (MHMS)	Deep neural networks with multi-layered nonlinearities are used by MHMS systems to derive more significant insights from health data.
Costa et al. [7] / Data management with IoHT	Through the easy collection and integration of vital sign data, this network seeks to enhance patient care in hospital environments.
Riazul Islam et al. [8]/ Medical Network Structures and Platforms	The authors propose different network structures and platforms within the healthcare context to facilitate data collection and transfer.
D Souza et al. [9]/ Fleck Nano Wireless Sensors and Accelerometers	The proposed solution utilizes these specific types of wireless sensors and accelerometers to track and monitor patients indoors. These sensors are likely used for collecting data on patient location and physical activities such as walking or running.
Kadhim et al. [10]/ Internet-based healthcare monitoring systems	Systematic reviews involve a structured and comprehensive approach to literature review, ensuring transparency, replicability, and minimizing bias.
Salatino et al. [12]/ Semantic Technologies and Machine Learning	Semantic technologies involve the use of semantic web standards to enhance data interoperability and understanding
Gope et al. [11]/ Fault-Tolerant Decision-Making Algorithms	They created fault-tolerant decision-making algorithms. These algorithms are designed to continue functioning effectively even in the presence of faults or errors, enhancing the reliability of decision-making processes.
Esmaeili et al. [13]/ Secure, Lightweight Sensing Approach for Body Area Networks	This approach involves the use of sensors in the context of a body area network to monitor various physiological parameters or health-related data. The emphasis is on ensuring security and efficiency in data sensing within the BAN framework.
Jamil et al. [14]/ Blockchain Technology and Smart Contracts/ Medical Sensors - Libelium E-health Biometric Sensor Platform Toolbox	They introduced a platform for tracking patient vital signs based on blockchain technology and smart contracts. The platform guarantees data integrity, privacy, and access control. These aspects are crucial for maintaining the security and confidentiality of patient data.
Dagher et al. [15]/ Blockchain-Based Framework - Ancile	They proposed Ancile, a blockchain-based framework designed for access control and interoperability. The framework utilizes a permissioned Ethereum Blockchain, which implies that the blockchain network requires permission for participation, typically ensuring a higher degree of control over participants.

Salatino et al. [12] employed semantic technologies and machine learning to investigate knowledge flow between academia and industry. Gope et al. [11] created fault-tolerant decision-making algorithms and authentication systems to build an Internet of Things-based healthcare system. A secure, lightweight sensing approach for body area networks was presented by Esmaeili et al. [13]. It identifies patient data according to a priority mechanism and provides emergency patients with services quickly. A platform for tracking patient vital signs that is based on blockchain technology and smart contracts was introduced by Jamil et al. [14]. Ancile, a Blockchain-based framework for access control and interoperability using a permissioned Ethereum Blockchain, was proposed by Dagher et al. [15]. To maintain many capabilities, including the consensus process, the tracking of data, permissions, re-encryption, node classification, and their relationship histories, its framework consists of multiple smart contracts.

III. IoT Sensors in Healthcare

Integrating intelligent systems efficiently is crucial for technical, safety, and economic reasons. Sensor devices, IoT, and reliable electrical installations can create comfortable environments that lead to cost reduction, increased productivity, and improved health and safety for people, companies, and institutions. Table 2 presents the IoT commonly used sensors in healthcare.

Table II. IoT Sensors

Sensor Type	Purpose	Example
Temperature Sensors	To identify fever or keep track of a patient's condition.	Infrared thermometers, thermocouples, thermistors.
Heart Rate Monitors	To determine your cardiac health and track physical activity, take a reading on your heart rate.	Electrocardiogram (ECG or EKG), and photoplethysmography (PPG) sensors in wearable devices.
Blood Pressure Monitors	To assess cardiovascular health, take your blood pressure.	Sphygmomanometers, automated blood pressure cuffs.
Blood Glucose Sensors	To assess cardiovascular health, take your blood pressure.	Continuous glucose monitors, and blood glucose meters.
Oxygen Saturation Sensors	Determine the blood's oxygen content.	Pulse oximeters.
Respiration Rate Sensors	Keep an eye on your breathing rate to evaluate your respiratory health.	Respiratory belts, spirometers.
Electroencephalogram (EEG) Sensors	For neurological evaluations, note brain electrical activity.	EEG electrodes.
Electromyography (EMG) Sensors	To evaluate the neuromuscular function of muscles, measure their electrical activity.	EMG electrodes
Accelerometers and Gyroscopes	To monitor physical activity and evaluate mobility, measure orientation and movement.	Inertial measurement units (IMUs) in wearable devices.
Medical Imaging Sensors	Take pictures to aid in diagnosis.	X-ray detectors, magnetic resonance imaging (MRI) sensors, and ultrasound transducers.
Chemical and Gas Sensors	Identify particular compounds or gases for environmental or diagnostic purposes.	Breath analysers, and environmental sensors in healthcare facilities.
Pressure Sensors	Measure pressure for use in a variety of medical settings.	Barometric sensors, and pressure transducers.

Energy audits in these human-occupied environments contribute to reduced greenhouse gas emissions, directly impacting human health, environmental protection, and profitability. These interconnected factors, affecting IoT platforms, energy control, and health, should be central to research on overall human health and well-being. The unstoppable advancements in IoT applications, along with sensors, intelligent devices, and advanced architectures, offer immense potential for development in cities and communities, both now and soon [23].

The rise of IoT and cloud technology has revolutionized communication, enabling seamless information sharing across devices. IoT acts as a stage to connect diverse gadgets, fostering collaboration and data preparation. This paves the way for innovative applications like smart homes, cities, and, most importantly, smart healthcare. IoT empowers objects to sense, measure, communicate, and share data through interconnected networks. These objects and sensors gather valuable information, analysed and utilized for tasks like classification, decision-making, and scheduling. With the emergence of 5G-based wireless, sensor, and IoT technologies, even remote healthcare for patients of all risk levels becomes readily available [5]. Sensitive healthcare information demands ironclad protection against unauthorized access, breaches, and misuse. This is where blockchain steps in, offering a promising solution. Unlike centralized systems, blockchain's decentralized nature distributes data across a network of interconnected nodes, making it incredibly resistant to manipulation. It's the perfect fortress for safeguarding sensitive healthcare data. Adding to its power, blockchain's inherent cryptography acts as an extra layer of security. This ensures authenticity and integrity, fostering trust and transparency within the healthcare ecosystem [16].

IV. Methodology

This study employed a mixed approach to investigate the potential of IoT systems and smart sensors to improve health outcomes and promote environmental sustainability. The rise of ML and AI is revolutionizing smart healthcare systems. These technologies excel at managing vast amounts of medical data, enabling low-latency and reliable analysis for improved outcomes. For instance, AI algorithms can sift through complex medical images to detect diseases like cancer at early stages, potentially saving lives. Similarly, ML models can continuously monitor patients' health data, identifying potential issues before they become critical, paving the way for personalized preventive care. Moreover, by understanding the dynamic nature of healthcare networks, ML can anticipate resource needs and optimize workflows, ensuring smoother operations. As AI becomes more integrated into medical units, the wealth of collected data can be transformed into actionable insights through advanced algorithms, facilitating predictive analytics and research that guide future healthcare advancements [5]. Fig. 2 shows the architecture of IoT in healthcare. It comprises network-connected sensors and medical devices, a cloud for storage of gathered information via these devices, and a subscriber in the form of monitoring the patient's information received.

Fig. 3 shows the role of IoT in healthcare. It helps in the early detection of diseases, improves decision-making, gives associated care, enhances user experience, and immediate attention to patients in case of emergencies.

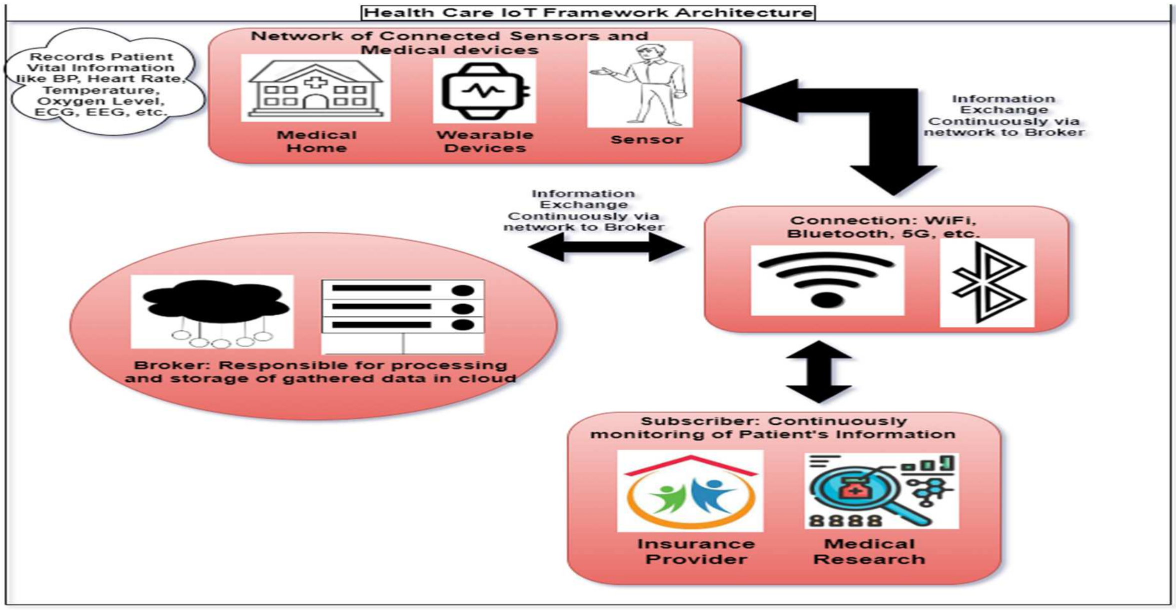

Fig. 2. Architecture of IoT in Healthcare

The Internet of Things (IoT) is revolutionizing the healthcare industry by making it possible for doctors to make better decisions, for patients to have a more individualized and connected experience, and for early disease diagnosis. IoT systems can offer real-time insights into a patient's health by gathering and evaluating data from wearables, sensors, and medical equipment. This enables early intervention and proactive care. Better health outcomes, lower expenditures, and a more effective healthcare system may result from this [25].

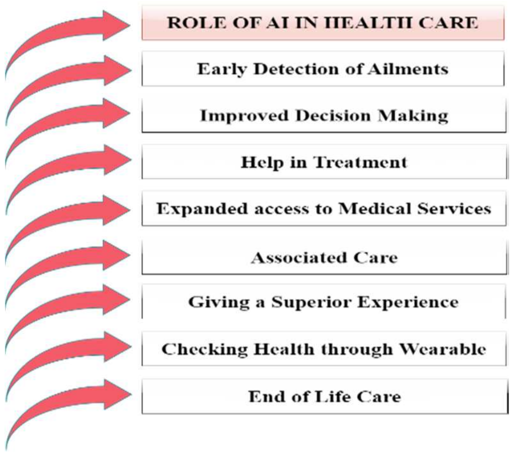

Fig. 3. Role of AI in Healthcare

Table 3 presents the application of IoT in healthcare. There are several ways in which the Internet of Things (IoT) is significantly influencing healthcare. IoT-enabled devices at hospitals, for instance, may track patients and equipment, keep an eye on vital signs, and even conduct surgery. IoT devices can help patients by reminding them to take their medications, monitoring their chronic diseases remotely, and even offering mental health care. IoT data can also give medical professionals insightful information about their patients' health and support them in choosing the best course of action. All things considered, IoT is improving healthcare by making it more effective, efficient, and customized.

TABLE III. APPLICATION OF IOT IN HEALTH CARE

IoT Applications in Health Care	Description
IoT in Smart Hospital	IoT-enabled hospitals are transforming patient safety and operational efficiency through Real-Time Location Systems (RTLS). Smart tags track assets, patients, and employees, boosting staff response times and equipment availability. Vulnerable populations are further protected by IoT hygiene devices that minimize infection rates by monitoring key areas.
IoT for Patients	IoT empowers patients with personalized healthcare through connected devices like fitness bands and blood pressure monitors. These wearables provide real-time health data, enabling doctors to tailor treatment plans and patients to actively participate in their own well-being. IoT seamlessly builds a comprehensive record of patients' health data, facilitating informed decisions and proactive care. This interconnectedness extends to emergencies: a medical alert bracelet automatically alerts designated contacts for instant help if a wearer falls, ensuring they receive immediate assistance when needed.
IoT for Physicians	IoT revolutionizes healthcare by enhancing doctor-patient interaction and enabling advanced treatment options. Through real-time data and remote monitoring, doctors can proactively manage patients' conditions and tailor care plans with precision. Micro-robots controlled via internet connections now perform intricate surgeries impossible for human hands, expanding surgical possibilities. Furthermore, IoT devices offer insights into emotional states by collecting data like heart rate and blood pressure, improving mental health monitoring and opening doors for novel therapeutic approaches.
IoT in Smart Monitoring of Labs	Beyond patient care, IoT revolutionizes laboratory work by safeguarding biosafety and optimizing workflows. By monitoring everything from temperature and air quality to equipment function, these connected devices ensure a secure environment for humans, experiments, and specimens. Real-time alerts for deviations in key parameters safeguard biosafety by triggering swift interventions and preventing accidents. Furthermore, connected lab devices streamline workflows, ensuring equipment operates flawlessly and protocols are meticulously followed, ultimately leading to faster research progress and reliable results.

V. CHALLENGES FACED

The healthcare sector has seen incredible technological advancement in recent years, with many applications of this technology to address healthcare-related problems. The availability of healthcare services at one's fingertips has greatly enhanced them. Smart sensors, cloud computing, and communication technologies have all been applied to transform the healthcare sector through the Internet of Things. IoT has certain problems and challenges, just like other technologies, which could open up new research avenues [26][27]. Fig. 4 shows some of the challenges that occurred in healthcare. Ensuring a more efficient and accurate healthcare system demands attention to numerous aspects, as highlighted by various studies. These are discussed below:

Patient engagement: Encouraging frequent use of medical gadgets empowers patients with data and fosters their active participation in managing their health.

Minimized response time: Real-time responsiveness through rapid data transmission and analysis is crucial for timely interventions and improved patient outcomes.

Reliable data communication: High-quality data transmission with minimal packet loss ensures accurate diagnoses and informed decision-making.

Leveraging ICT effectively: Understanding the symbiotic relationship between Information and Communication Technologies (ICT) and healthcare is vital for optimizing system design and maximizing benefits.

Software and device integrity: Prioritizing software safety, reliability, and secure device connectivity is paramount for patient safety and data protection. Optimizing network architecture further enhances communication efficiency and system performance.

Early detection capabilities: Advanced monitoring systems with the ability to detect diseases like hypertension at an early stage can significantly improve patient prognoses.

Equitable access to care: Ensuring healthcare services are accessible and equitable, particularly in remote and developing areas, is essential for achieving universal healthcare goals.

Continuous technological adaptation: Embracing the latest advancements in healthcare technology allows continuous improvement and delivery of the best possible care.

Education and workforce development: Establishing targeted educational pathways in smart healthcare engineering can cultivate a skilled workforce prepared to contribute to this rapidly evolving field.

Building upon the core requirements, one must also consider crucial technical aspects for optimized performance and sustainability of IoT usage in healthcare. Minimizing battery drain and addressing potential leakage in energy storage devices are paramount for long-term functionality. Optimizing hardware and software for low-power operation ensures system stability in resource-constrained environments. Efficiently powering IoT terminal nodes, the foundation for smart decision-making and data processing, requires careful consideration of communication protocols, data storage strategies, and overall system architecture. By prioritizing these technical considerations, one can enhance communication efficiency, secure reliable data storage, and ultimately deliver a robust and sustainable smart healthcare system [17][28].

Fig. 4. Challenges Faced in Healthcare

VI. FUTURE DEVELOPMENTS OF IOT IN HEALTHCARE

Future developments in IoT for healthcare could bring about revolutionary changes by allowing the seamless integration of smart, networked devices and technologies (Fig. 5). The convergence of IoT and healthcare is anticipated to improve overall healthcare outcomes, expedite medical procedures, and improve patient care going forward [29]. More advanced remote patient monitoring will make it possible to track vital signs and other health indicators in real time, giving medical personnel the ability to intervene quickly and customize treatment regimens [30]. The integration of wearable technology, sensors, and intelligent medical equipment will not only enable uninterrupted monitoring but also enable patients to take an active role in overseeing their health.

Fig. 5. Future Concerns of IoT

Furthermore, IoT-driven data analytics will be essential in helping to extract valuable insights from the massive amounts of generated health-related data, which will support preventive measures, predictive healthcare models, and more efficient decision-making [31]. Security protocols and interoperability standards will also be essential focal points to guarantee the smooth exchange of data while protecting patient privacy. In the end, IoT in healthcare has enormous potential to build a more integrated, effective, and patient-focused healthcare ecosystem [32].

CONCLUSION

IoT is being used in practically every aspect of human endeavor and is expected to be a major factor in the future development of the Internet. Over the course of the next ten years, billions of IoT devices and smart applications will appear in several domains (viz. augmented reality, smart manufacturing, smart agriculture, smart healthcare, and more). A strong, dependable, and adaptable IoT architecture to optimize network systems is necessary and urgent to meet these enormous demands. The current review looked at a few aspects of the healthcare IoT system. The paper provides information on the healthcare services that are currently offered. By utilizing these ideas, IoT technology has assisted medical practitioners in measuring a wide range of health parameters, diagnosing and monitoring multiple health issues, and providing diagnostic services at remote locations. It has changed the focus of the healthcare sector from being primarily hospital-centric to being more patient-centric. The paper also presents many uses of the IoT and their most recent developments. Additionally, the difficulties and problems of the healthcare IoT system's conception, production, and application have been detailed.

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